SARDAR UMAIR NAWAZ KHAN

AI Systems Engineer | Machine Learning & Agentic Automation

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PROFESSIONAL SUMMARY

Computer Science undergraduate at FAST-NUCES specializing in Agentic AI systems, Machine Learning architectures, and high-performance backend pipelines. Experienced in training predictive models and neural pipelines with PyTorch and Scikit-learn, engineering autonomous multi-tool LLM workflows with LangChain and n8n, designing asynchronous RESTful APIs using FastAPI, Node.js, and PHP, and implementing normalized relational databases in PostgreSQL and MySQL. Hands-on experience architecting end-to-end full-stack AI applications, resilient automation workflows, and scalable backend APIs.

TECHNICAL SKILLS

Programming Languages: Python, Modern C++ (C++17), TypeScript, JavaScript (ES6+), PHP, SQL, Bash

AI, Generative AI & ML: PyTorch, Scikit-learn, Generative AI, LLM Integrations, Fine-Tuning, Autonomous AI Agents, LangChain, n8n Orchestration, RAG Workflows, PCA, Hierarchical Clustering, SMOTE

Backend & Systems: FastAPI (Async Python), Node.js, Express.js, PHP, React 18, Vite, Asynchronous I/O (asyncpg), RESTful APIs, Session Security (Scrypt, OTP)

Databases & DevOps: PostgreSQL, MySQL, Relational Database Architecture, Docker, Linux, Git, GitHub, Postman

TECHNICAL PROJECTS

AI Appointment Assistant & Workflow Automation Engine **[GitHub Code]**

FastAPI, React 18, PostgreSQL, n8n, Google Gemini, Docker

- Architected a modular multi-tenant appointment architecture with containerized Docker deployment, combining a React 18 chat portal, asynchronous FastAPI backend, and autonomous n8n workflow powered by Google Gemini.
- Engineered function-calling tools for real-time slot checking and conflict-free database commits; implemented dynamic RFC-5545 `.ics` calendar invite generation and dual-engine SQL fallback resilience.

Autonomous AI Meeting Scheduling Agent **[GitHub Code]**

Python, n8n, Zoom API, Google Calendar API, Gmail API, Docker

- Engineered an autonomous meeting orchestration pipeline using n8n to ingest webhook payloads, negotiate time slots, and provision video conferences.
- Integrated Zoom Server-to-Server OAuth, Google Calendar, and Gmail APIs to automate room creation, invite dispatch, and calendar synchronization.

Customer Segmentation via Hierarchical Clustering **[GitHub Code]**

Python, Scikit-learn, SciPy, Agglomerative Clustering, PCA, Pandas

- Processed 541,000+ transactional line items into RFM customer behavioral feature vectors across 4,372 accounts, applying feature scaling and 2-component PCA (97.39% variance explained).
- Implemented Agglomerative Hierarchical Clustering using Ward's minimum-variance linkage and SciPy dendrograms to classify users into 3 distinct behavioral cohorts.

Financial Fraud Detection & Anomaly Classification System **[GitHub Code]**

Python, Scikit-learn, Imbalanced-Learn, SMOTE, Pandas

- Built an end-to-end anomaly detection pipeline on 284,000+ financial transactions with acute class skew (0.17% fraud prevalence).
- Standardized 30 high-dimensional feature vectors, applied SMOTE to balance minority fraud representation without data leakage, and achieved a **90.82% fraud interception recall rate** with 97.73% overall accuracy.

Full-Stack Hostel Management System **[GitHub Code]**

Node.js, Express.js, MySQL, AWS RDS, Modular MVC

- Developed a full-stack student housing operations platform managing rooms, resident allocations, staff administration, and fee tracking.
- Architected a normalized 25-table relational schema deployed on AWS RDS with strict foreign keys, query indexing, and modular RESTful API endpoints.

International Banking Management System **[GitHub Code]**

Modern C++17, OOP Architecture, File Stream Serialization, Data Persistence

- Architected a high-integrity banking engine in modern C++17 utilizing memory-safe OOP design patterns, runtime polymorphism, and strict state encapsulation.
- Implemented credential authentication routines, defensive input validation, atomic balance transfers, and low-level file stream persistence via CSV serialization.

EDUCATION

FAST National University of Computer and Emerging Sciences (FAST-NUCES) Peshawar, Pakistan

Bachelor of Science in Computer Science (BSCS) **Aug 2024 – Jun 2028 (Expected)**

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (C++), Operating Systems, Database Systems

Government Sadiq Abbas Postgraduate College | *HSSC (FSc Pre-Engineering)* Dera Nawab Sahib, Pakistan | **2022 – 2024**

EXTRACURRICULARS & PROBLEM SOLVING

Competitive Chess (@umair-nawaz): Active rapid player with **4,500+ tactical puzzles solved**; passionate about algorithmic game theory and combinatorial problem-solving.